



SUPPORT HUB

A modern Android-based enterprise helpdesk and ticket management application designed to streamline organizational support operations.

Built for the CodeMate Community

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01. PROJECT AT A GLANCE

SupportHub is a **modern Android-based enterprise helpdesk** designed to streamline organizational support operations. It provides dedicated experiences for Employees, Agents, and Administrators, enabling real-time ticket tracking and resolution.

By utilizing Kotlin and Firebase, SupportHub ensures high performance and reliable data synchronization across all user roles within a centralized environment.

Core Modules

Module	Scope
Employee Portal	Interface for creating, tracking, and searching support requests with personal history.
Agent Portal	Productivity dashboard for viewing assigned tickets, updating status, and resolving issues.
Admin Portal	Management suite for user approvals, role assignments, and ticket allocation control.
Ticket Management	Lifecycle management from creation and assignment to status updates and resolution.
Real-time Sync	Firestore-driven updates for instant status changes and communication logs.

Module	Scope
Search & Filters	Global retrieval system for tickets, employees, and agents across the workspace.
Profile & Settings	Role-specific configuration for account details and application preferences.
Workspace Ops	Organization-wide support tracking and multi-role operational monitoring.

02. PROBLEM, USERS & PRODUCT VALUE

The Problem

Organizations often manage internal support requests through fragmented emails, phone calls, and spreadsheets. This leads to poor ticket tracking, a lack of accountability, and difficult communication between departments.

Target Users

- **Employee:** Creates and tracks support tickets while monitoring resolution progress in real-time.
- **Support Agent:** Views assigned ticket queues, updates status, and actively resolves technical issues.
- **Administrator:** Approves users, assigns organizational roles, and manages overall ticket allocation.

Product Value Proposition

The platform delivers value through a cohesive enterprise strategy:

03. CUSTOMER PERSONA

Primary Persona: Admin / Support Manager

The primary persona is an Organization Administrator or Support Manager responsible for overseeing organizational support operations from a centralized mobile platform.

Goals & Needs

- **User Management:** Efficiently approve new users and assign correct roles within the system.
- **Workflow Visibility:** Maintain full oversight of ticket progress and support team performance.
- **Resource Allocation:** Rapidly allocate incoming tickets to available agents to reduce response times.

Pain Points

- **Manual Handling:** Tedious ticket processing via spreadsheets and scattered communications.
- **Blind Spots:** Difficulty monitoring real-time ticket updates and employee satisfaction.

04. ARCHITECTURE & TECHNOLOGY

SupportHub utilizes a modern Android architecture centered around Jetpack Compose and Firebase for scalable enterprise performance.

Technology Stack

Layer	Technology
Frontend	Kotlin, Jetpack Compose, Material 3
Architecture	MVVM, Repository Pattern, StateFlow
Database	Cloud Firestore (Real-time synchronization)
Authentication	Firebase Auth (Google Sign-In + Email/Pass)
Async Ops	Kotlin Coroutines
Navigation	Navigation Compose
Infrastructure	Firebase Services

Architecture Flow

The system maintains a clean request-response lifecycle to ensure data integrity and real-time responsiveness:

User → Android App → ViewModel → Repository → Firebase Auth / Firestore

05. CORE SUPPORT WORKFLOW

The workflow is designed around the lifecycle of a support ticket, ensuring seamless communication between employees, admins, and agents.

Operational Lifecycle

1. **Ticket Creation:** Employees submit support requests via a dedicated portal.
2. **Admin Review:** Administrators verify incoming tickets and approve user registrations.
3. **Agent Allocation:** Admins assign specific tickets to qualified support agents.
4. **Issue Resolution:** Agents update status from Open to In Progress to Resolved.
5. **Real-time Update:** Employees receive instant notifications of resolution via Firebase.



06. FEATURE MAP

Feature	Description
Auth & Roles	Role-based access control for Employees, Agents, and Administrators.
Employee Panel	Create, search, and track support tickets with full status history.
Agent Panel	View assigned ticket queue, update status, and resolve issues.
Admin Panel	Management of user approvals, role assignments, and ticket allocation.
Notifications	Real-time updates for ticket creation, assignments, and resolution status.
Global Search	Fast filtering for tickets, customers, and team members across the app.
Knowledge Base	Access to FAQs and documentation for employee self-service support.
Chat & Connect	Real-time communication logs between employees and assigned agents.

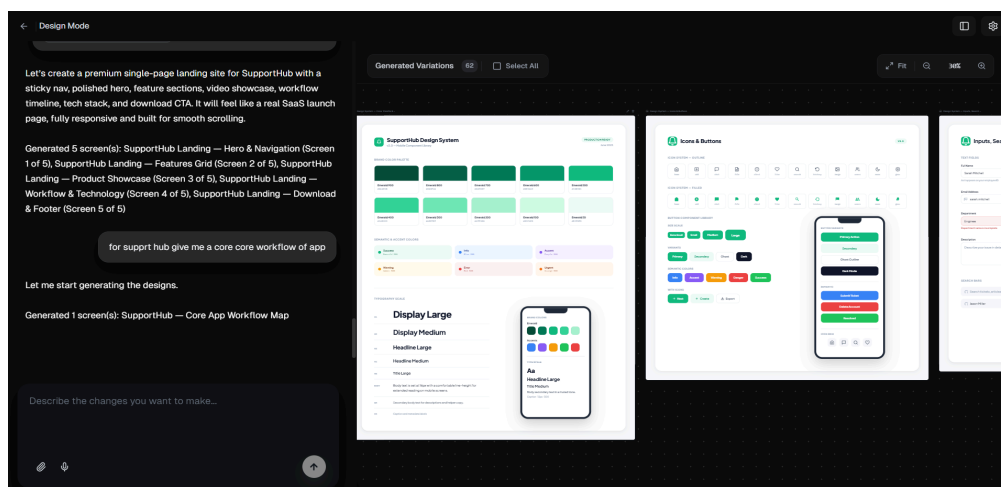
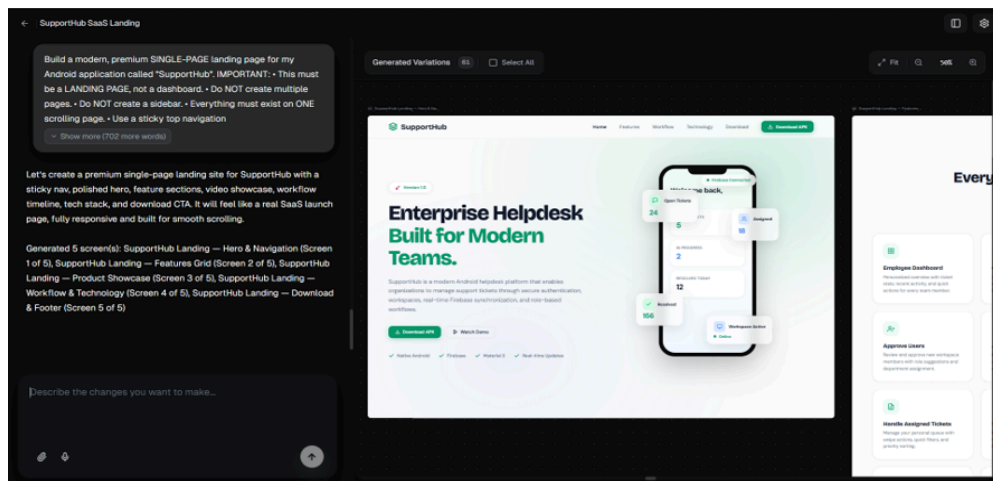
07. BUILD: DESIGN MODE

Design Mode served as the environment for **visual identity creation and system mapping**. It established the Material 3 enterprise look-and-feel across over 50 screens.

Design Workflow

Concept → Material 3 System → Portal Mockups → Review → Refinement

Design Mode utilized Behance-style mobile UI showcases to display realistic phone mockups for authentication, employee, agent, and admin portals.



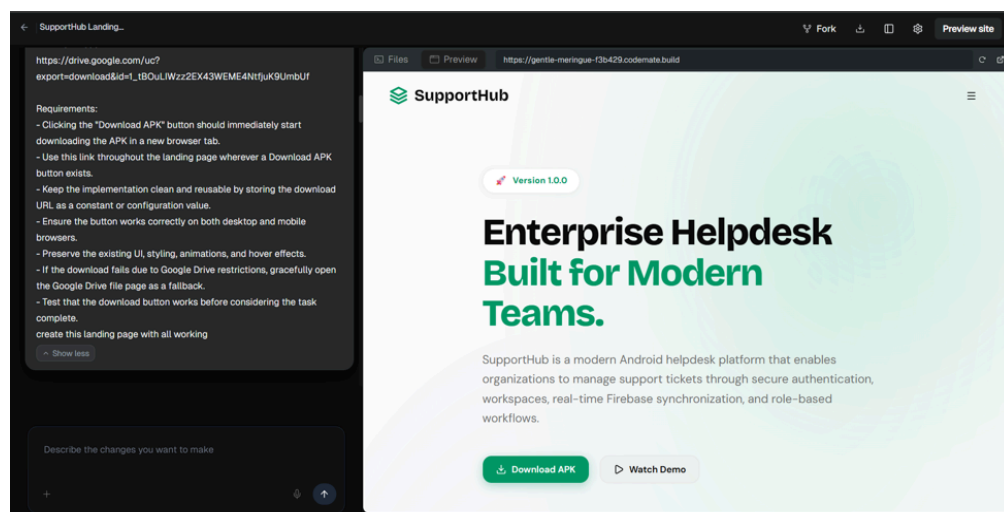
08. BUILD: PROTOTYPE MODE

Code Mode transitioned finalized designs into a functional landing page with production-ready code and demo integration.

Implementation Strategy

The build focused on creating a responsive product landing page with hero sections, workflows, and APK download functionality.

- **Prompt 1 : Landing Page Conversion** - Build a complete SupportHub landing page using the approved designs already created. Convert designs into a fully responsive, production-ready website with smooth animations and a premium UI across desktop, tablet, and mobile.
- **Prompt 2 : Product Workflow Animation** - Create a beautiful animated product workflow showing the complete support process from User Login to Agent Resolution. Present each step clearly with smooth scrolling animations and a professional enterprise helpdesk style.
- **Prompt 3 : Tech & Download Section** - Add a dedicated technology section showcasing Kotlin, Jetpack Compose, and Firebase. Follow this with a premium download section containing Android platform info, APK download buttons, and project links.
- **Prompt 4 : Functional Verification** - Ensure every button, animation, navigation element, and video preview works correctly. Add smooth transitions and hover effects to create a polished SaaS-ready landing page experience.



09. BUILD PROMPT HIGHLIGHTS

The build workflow prioritized creating a premium Material 3 experience with organization-specific controls.

Representative Prompts

1. **Design Foundation:** Design a premium Behance-style mobile UI showcase for the App Launch and Authentication flow of "SupportHub". Use realistic Android phone mockups with professional CTA buttons and elegant enterprise themes, incorporating distinct onboarding paths for new and returning users, clear error handling states, and streamlined single sign-on options.
2. **Employee Experience:** Create a complete Employee experience for SupportHub including dashboard, ticket creation, and tracking status. Keep the interface intuitive and focused on making support requests simple to create and monitor, featuring real-time status notifications, quick-action ticket templates, and comprehensive submission history views.
3. **Functional Landing Page:** Build a professional SupportHub landing page that introduces the application, explains the support workflow, and provides a clear APK download experience with smooth navigation and premium animations, complete with interactive feature showcases, system requirement details, and responsive layouts across mobile, tablet, and desktop views.

10. CORA: PLANNING & IMPLEMENTATION

CORA served as the primary implementation layer, handling architecture planning, Firebase synchronization, and navigation logic fixes.

Implementation Workflow

Requirement → CORA Architecture → Implementation → Testing → UI Polish

Strategic Prompts

- **Prompt 1 : Architecture Planning** - Analyze the SupportHub codebase. Identify relevant modules, database models, and API endpoints for role-based dashboard logic. Create a structured plan for implementing separate Employee, Agent, and Admin experiences without breaking existing patterns.
- **Prompt 2 : Firebase & Auth Integration** - Connect SupportHub with Firebase services using google-services.json. Configure Auth and Firestore operations to ensure user profiles are retrieved correctly and real-time synchronization functions across the workspace.
- **Prompt 3 : Ticket Management Refinement** - Build the Employee Ticket Management module. Use Firestore snapshot listeners to ensure status changes and resolution notes appear automatically for the user. Optimize queries and implement offline error states.
- **Prompt 4 : Navigation & Dashboard Logic** - Review the onboarding flow. Check approvedRole before deciding where a user should navigate. Prevent duplicate navigation events and ensure pending users are restricted to the Approval screen.

Working (3m 54s) ▾

Awaiting reference screenshot required by the new scope before any implementation. The screenshot is now the primary source of truth, so code changes must pause until the image is available for direct analysis.

⚡ Checkpoint

Updating todos 5 items

Blocked on the required reference image. The new task explicitly makes the screenshot the source of truth, but the image was not included in this message yet, so pixel-accurate implementation cannot start without violating the "do not estimate" constraint.

Using a tool while waiting for the promised screenshot. Inspecting current employee tickets screen implementation so work can begin immediately after image arrival.

⚡ Checkpoint

Read 3 files EmployeeTicketsScreen.kt, TicketViewModel.kt ... >

Current codebase assessment completed with tool evidence:

- `EmployeeTicketsScreen()` is currently just a placeholder wrapper around `EmployeePageTemplate()`.
- Ticket data already exists through `TicketViewModel`, with live ticket observation in `start()`.
- Existing tests currently cover ticket submission/viewmodel

☰ Todos 5 / 5 ▾

Write or edit code...

<> Code

⦿ Standard



I have attached a reference screenshot. ****Analyze the screenshot pixel-by-pixel**** and create the UI as closely as possible. Do ****not**** estimate the design—extract every visual detail from the image.

Objective

Redesign the Employee → Tickets** screen to match the uploaded screenshot while keeping it production-ready and enterprise-grade....



Working (3m 34s) ▾

Awaiting reference screenshot required by the new scope before any implementation. The screenshot is now the primary source of truth, so code changes must pause until the image is available for direct analysis.

⚡ Checkpoint

☰ Todos 5/5 ▾

Write or edit code...

<> Code

⦿ Standard



11. CORA TASK ARCHIVE

Category	Implementation Focus
Authentication	Email/Pass, Google Sign-In, and role-based redirect logic.
Cloud Firestore	User data, tickets, notifications, and real-time snapshot listeners.
Ticket Lifecycle	Creation, assignment, status tracking, and priority handling.
Dashboard System	Live statistics, assigned ticket lists, and role-specific KPI cards.
Profile & Settings	Management of account details, images, and theme preferences.
UI/UX Refinement	Jetpack Compose screens, Material 3 styling, and animations.

12. IMPLEMENTATION MILESTONES

Phase	Milestone	Focus Area
1	Planning	ITSM concept definition and Firebase architectural structure.
2	Design System	Portal mockups for Auth, Employee, Agent, and Admin roles.
3	Core App Dev	Kotlin, Jetpack Compose, and MVVM infrastructure setup.
4	Firebase Setup	Authentication and real-time Firestore database integration.
5	Workflow Dev	Role-based dashboards and ticket lifecycle implementation.
6	Landing Page	Responsive SaaS site with APK download and demo videos.

13. BACKEND, DATA & INTEGRATIONS

SupportHub leverages Firebase as a unified backend service to maintain organizational data integrity and real-time availability across all Android portals.

Core Services

- **Firestore Auth:** Manages secure user entry points including standard email credentials and Google Sign-In for enterprise onboarding.
- **Cloud Firestore:** Acts as the primary NoSQL data store, handling synchronization for tickets, user roles, and workspace-specific settings with sub-second latency.
- **Firestore Storage:** Handles binary asset management for ticket attachments, profile images, and organizational documentation storage.

Data Collections

The database utilizes a structured collection hierarchy: Users (profiles and roles), Workspaces (org details), and Tickets (support requests and audit logs). Every transaction is scoped to the organization identifier to ensure multi-tenant data isolation.

(default) ⓘ + Start collection departments messages notifications tickets users > workspaces	users + Add document 0An4JfbBofxjEkLuf0Re BwPQKgewJgbKuQWK0nqLckJkuzx2 > VBbL1jl96KP5ninpi4YCRKjW0Bd2	BwPQKgewJgbKuQWK0nqLckJkuzx2 + Start collection + Add field approvedRole: "admin" authProvider: "google" createdAt: 1784279518274 email: "swsdhindwibedy@gmail.com" fullName: "Swadhin Dwibedy" lastLogin: 1784293976747 requestedRole: "admin" status: "active" uid: "BwPQKgewJgbKuQWK0nqLckJkuzx2" updatedAt: 1784279518274 workspaceName: "swadhin" workspaceOwner: true
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SupportHub

Cloud Firestore

Database

(default)

Ask Gemini about the core concepts of using Firestore

DataRulesIndexesDisaster recoveryUsageQuery InsightsExtensions

More in Google Cloud

Panel viewQuery builder

workspacescodemate

(default)

+ Start collection

conversations

departments

notifications

tickets

users

workspaces

workspaces

+ Add document

SupportHub Demo

abc company

bis001

codemate

i00aaaa

swadhin

vAPsQbrLUMFLMPUMKGqc

codemate

+ Start collection

+ Add field

createdAt: 1784972299177

normalizedWorkspaceKey: "codemate"

ownerUid: "WkQISiqX63gKKLoqXeLXc6kP8Qh1"

status: "active"

updatedAt: 1784972299177

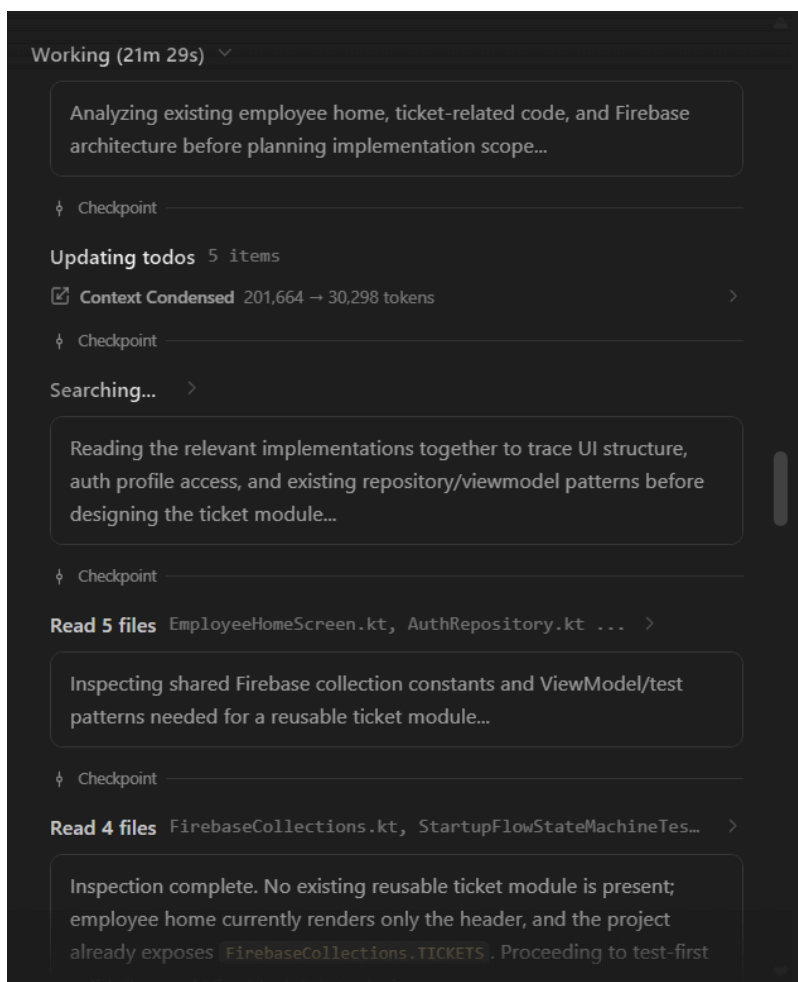
workspaceName: "CodeMate"

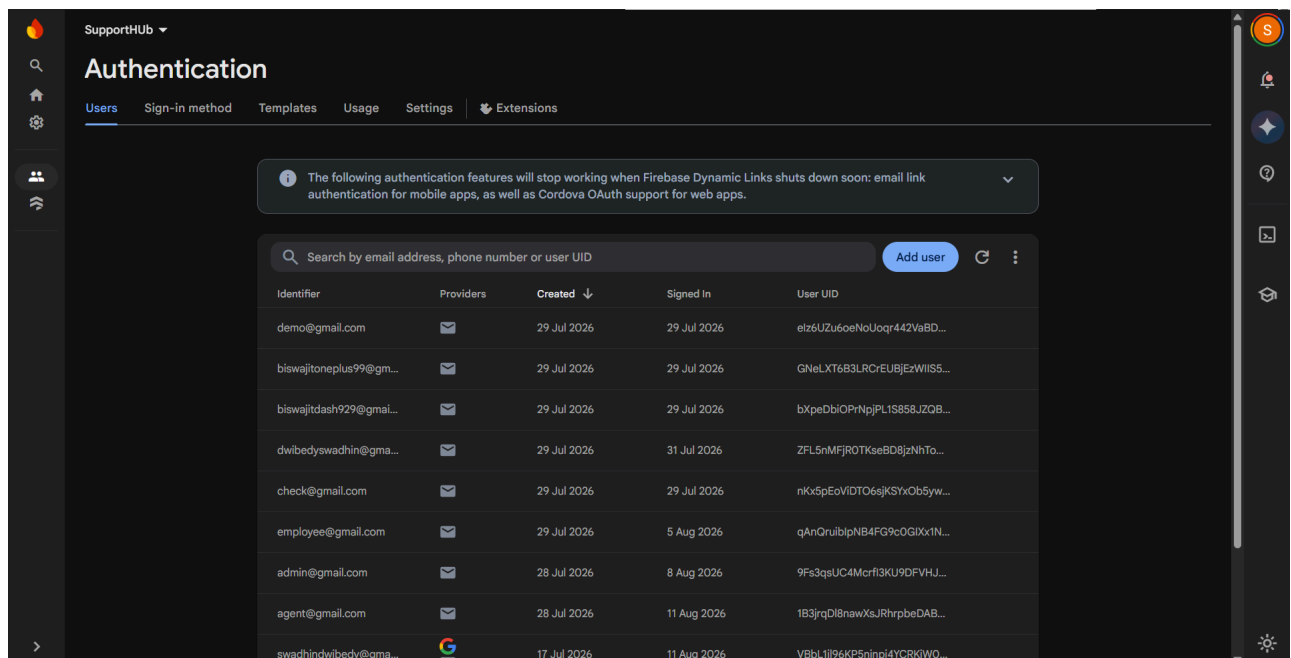
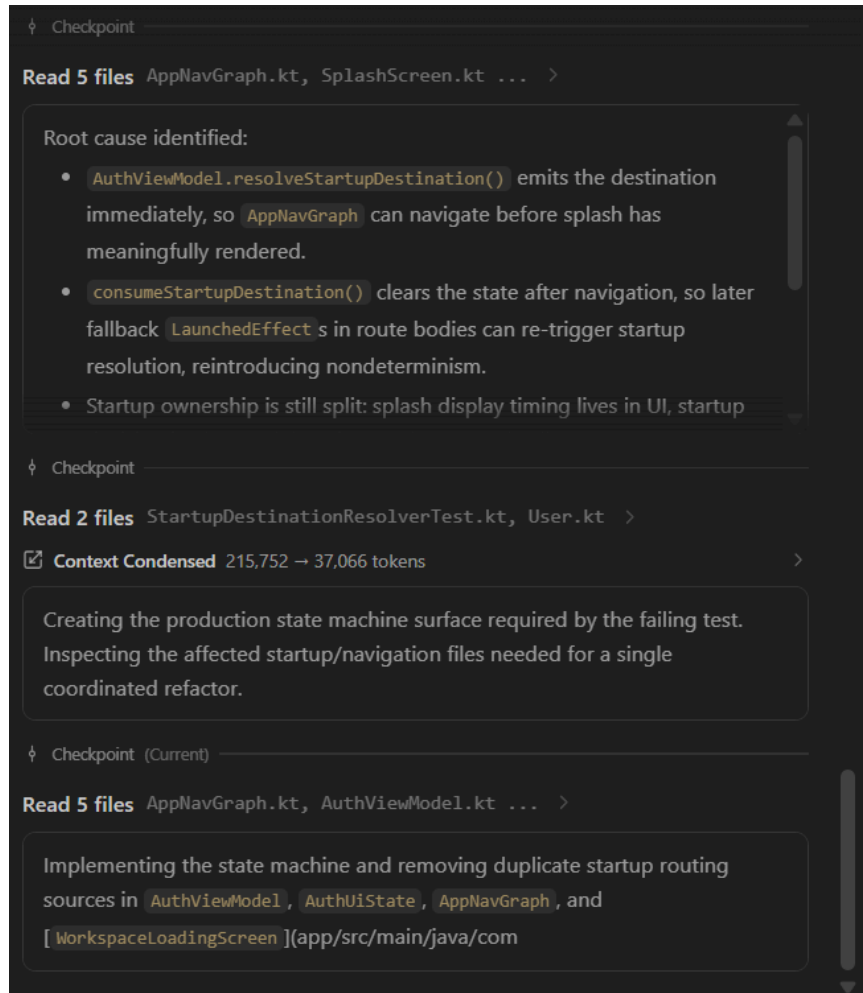
14. AUTHENTICATION & USER MANAGEMENT

SupportHub uses role-based access control (RBAC) to redirect users to the appropriate operational workspace after validation.

Role Capabilities

- **Employee Role:** Authorized to submit requests, track ticket status, and use self-service knowledge tools.
- **Agent Role:** Permissioned to manage assigned queues, communicate with customers, and resolve tickets.
- **Admin Role:** System-level access for approving users, assigning organizational roles, and monitoring analytics.

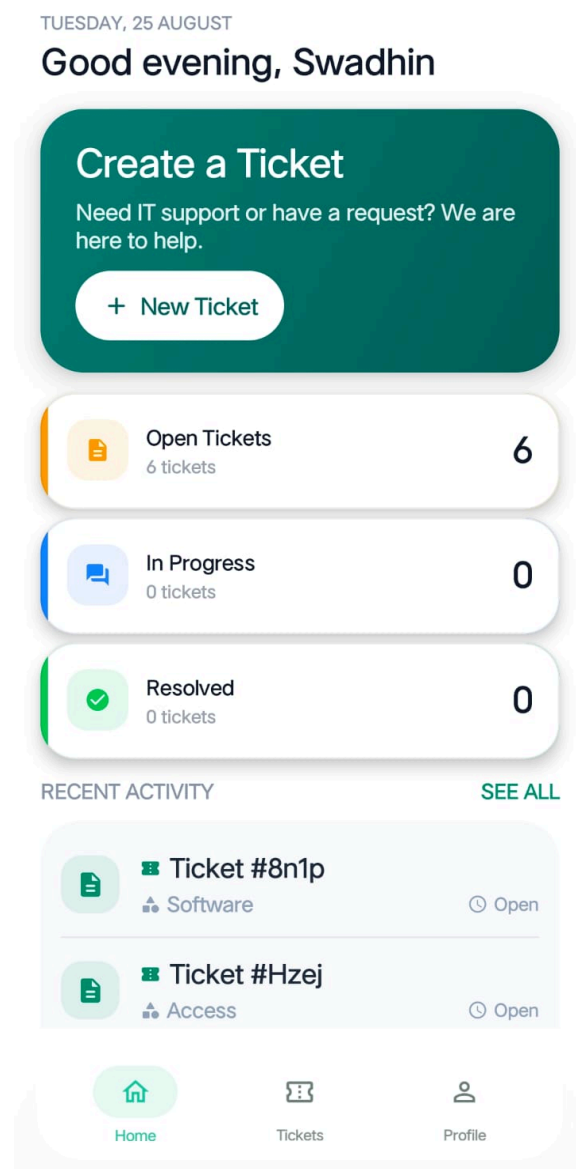




15. RESPONSIVE UI & POLISH

SupportHub is built with a **Material 3 First** principle to ensure a premium enterprise experience across all mobile and landing page layouts.

Detailed visual representations for the platform and landing page responsive designs are provided below.



GOOD AFTERNOON

Demo Agent

You have 7 active tickets waiting.

06:20 PM

Online



7

Assigned Tickets
+3 today



1

In Progress
Updated just now



3

Resolved Today
Last refreshed 2 min ...



0

Pending
4 waiting

Quick Actions



Queue



My Tickets



Home



Queue



Perf



Profile

Quick Actions



Queue



My Tickets



Resolved



Refresh

Recent Tickets

[View All](#)

Latest assigned tickets

 dQKe

RESOLVED

auth bug

demo Employee

3 Aug • 04:18 PM >

 wPwE

RESOLVED

email error

demo Employee

30 Jul • 12:07 PM >

 YDFB

RESOLVED

application issue



Home



Queue



Perf



Profile

Users

Refresh

Manage everyone in your works...

Total...
9

Pen...
0

Active
9

Adm...
2

Search users

- All
- Employees
- Agents
- Admi

Users

Refresh

AS

Aditya Suman Mathan

aditya.mathan@codemate.ai

Employee

>

AA

Admin

Admin@gmail.com

Admin engineering

>

ADMIN

Home

Users

Tickets

Profile